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Pre-Final Report

Project Name: V-FAB 1.0: A Multiscale Digital Twin of Semiconductor Processes in a Foundry

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1. Executive Summary

1.1 Project Overview

This project involved developing an interactive virtual cleanroom to help students and engineers understand semiconductor manufacturing processes through hands-on simulation. Traditional semiconductor education faces practical barriers: physical cleanrooms have strict access restrictions, specialized equipment is expensive to maintain, and chemical handling requires rigorous safety protocols. As a result, students rarely get opportunities for experimentation with the parameters presented to them.

Our team developed an interactive 3D simulation of the photolithography workflow using Unity. Rather than automating the manufacturing sequence into a single push-button process, the project intentionally focuses on manual step-by-step user interaction. Users navigate the laboratory, operate individual workstations, handle wafer carriers and the silicon wafer, adjust process parameters, and observe the resulting physical and structural changes. And this simulation is powered by a backend pipeline of deep machine learning surrogate models that compute optical diffraction and chemical diffusion in real-time.

1.2 Laboratory Environment and System Design

Development began with simple cleanroom layouts and placeholder geometry, which gradually evolved into a detailed modular laboratory. Built in Unity, the environment includes process benches, chemical fume hoods, appropriate machinery, cleanroom surface materials, and appropriate ambient lighting. To reflect real-world lab standards, the photolithography area uses yellow lighting similar to how it is used in physical fabs to prevent accidental photoresist exposure.

The simulation relies on first-person spatial navigation to build procedural understanding. Users move through the lab and interact with workstations by physical proximity and direct line-of-sight.

1.3 Equipment modelling and Animation

A key area of development was modeling and animating the semiconductor processing equipment. Standard equipment such as Spincoater, Hotplate etc were incorporated into the scene. The models were equipped with interactive controls, digital readouts, and automated dispense nozzles. The primary mechanical focus was the mask aligner station.

Because open-source 3D models of industrial alignment equipment were unavailable, An EVG Mask aligner machine was modelled from scratch in Blender using technical documentation, specifications, and reference videos of industrial EVG systems. The model includes internal and

external components such as tubing, wiring, microscope tracks, vacuum chucks, and motorized photomask trays. In Unity, Kinematic animations were also developed for automated tray ejection and insertion, photomask clamping, wafer leveling, and exposure operations.

1.4 Process & Chemical Visualization

To connect mechanical operations with material behavior, we added visual feedback for physical and chemical changes across the workstations. During spin coating, fluid deformation meshes and particle systems simulate liquid photoresist being deposited onto the spinning wafer.

During the soft baking step, a transition in color shaders on top of the wafer communicate the completion of the steps with most of the solvent molecules evaporated.

A detailed print of the photomask pattern had been planned to display after development process as well but was scrapped due to the possibility of being misleading.

1.5 Backend architecture & Machine Learning Integration

To support realistic experimentation without relying on expensive cloud computing servers, the simulation relies on a fully localized machine learning backend. Building this architecture required bridging offline physical simulations with real-time game engine rendering. The development of this backend followed a strict three-phase pipeline:

- Data Generation & Physical Simulation: Before any machine learning could occur, rigorous physical simulations were required to establish ground truth data. We used Python to completely generate these datasets from scratch making use of proper rigorous physics and chemistry equations. This provided the exact mathematical targets for how 193nm DUV light diffracts through a mask and how photoacid diffuses during baking.
- Surrogate Model Training (PyTorch): Using this synthetic data, custom neural networks were constructed and trained in PyTorch. For spatial, image-to-image transformations (like the optical exposure and chemical diffusion), multi-channel Convolutional U-Net architectures were deployed. For parameter-driven, non-spatial tasks (like spin coating fluid dynamics), Multi-Layer Perceptrons (MLPs) were utilized. These surrogate models successfully learned to approximate heavy PDE physics in a fraction of a second.
- The ONNX to Unity Sentis Bridge: To bring these Python-trained models into the interactive simulation, the trained weights were exported to the Open Neural Network Exchange (ONNX) format. Unity's Sentis neural network inference library was integrated to natively ingest these ONNX models. This critical bridge allows the user's local GPU, even running inside a WebGL browser environment, to execute

multidimensional tensor arrays (up to 512×512 spatial resolution) in real-time without external API calls.

Each workstation feeds user-selected parameters into the Sentis models:

SC1 & HMDS priming: Uses standard RCA SC1 cleaning parameters alongside user-defined HMDS prime temperature and duration to calculate physical undercut etch depth and the resulting surface contact angle for optimal photoresist adhesion.

Spin Coater Engine: Uses rotational velocity (RPM), dispense time, and fluid viscosity to calculate a 256-sample radial thickness and coverage profile across the wafer.

Soft Bake & PEB Engines: Use bake temperature and duration to calculate solvent evaporation rates and acid diffusion heatmaps.

Lithography Aerial Engine: Evaluates optical settings such as exposure dose (mJ/cm2), defocus (μm), numerical aperture (NA), to simulate aerial image intensity across the photomask pattern.

Chemical Resist Engine: Uses the aerial intensity map and critical dimension (CD) to calculate a 2-channel chemical resist profile, which is passed to the development station to determine final dissolution depth and feature dimensions.

This integrated pipeline allows users to see how adjustments to parameters like focus or spin speed directly impact feature fidelity and cause typical lithography defects.

1.6 Workflow Guidance & Accessibility

The available selection of UI includes Spincoater, Hotplate, Photomask tray, Mask Aligner hardware and software controls and the Development station controls.

To make the simulation approachable for first-time users, we implemented a gamified simplification of UI and controls, specifically limiting it to sliders, and buttons for accessibility without compromising much on realistic control. The Mask Aligner software, due to its importance and standard operating manual available online specifically underwent a close recreation of the real UI. However, a few details were omitted for the sake of simplicity.

A centralized workflow manager guides users through a structured (roughly) 60-step recipe covering the entire lithography process. To prevent visual clutter, we implemented contextual spatial highlighting: interactive wafer carrier, machine buttons, and placement docks highlight softly only when they are the active target of the current procedural step. Once an action is completed, the highlight turns off.

1.7 WebGL Deployment and Performance Optimizations

The final phase of development focused on deploying the application as a Unity WebGL experience that could run directly inside a web browser. The website 'Itch.io' was the perfect intermediary candidate to deploy the game and test it in real time. Since the simulation combines real-time 3D rendering with machine learning inference, particular attention was given to memory usage, browser stability, and compatibility with lower-end hardware.

Several optimizations were implemented to improve performance and accessibility, including memory tuning, build compression, and automatic fallback mechanisms for systems with limited graphics capabilities. As a result, the application was successfully made deployable on modern browsers while remaining functional on standard laptops and systems using integrated graphics hardware.

1.8 Summary Conclusion

The V-FAB platform provides an accessible, web-based simulation that connects theoretical photolithography principles with practical laboratory operation. By combining detailed mechanical modeling, step-by-step interactive workflows, real-time machine learning inference, and optimized WebGL deployment, the project delivers a functional and reliable educational tool for semiconductor manufacturing.

The game is accessible via this [Game Link] and its [Github Repo]

2. INTRODUCTION & BACKGROUND

Teaching semiconductor fabrication in an academic setting has traditionally relied on theoretical textbooks, 2D diagrams, and passive video demonstrations. Practical cleanroom access is often restricted due to high operational costs, contamination risks, and safety hazards associated with toxic chemicals and UV radiation.

Previous attempts have been made to virtualize semiconductor education, notably vfablab.org web platform. vfablab.org website provides fixed procedural walkthroughs that do not allow students to freely modify process parameters and observe real-time physical consequences. This project addresses these educational gaps by delivering an interactive, physics-driven 3D laboratory where students can actively manipulate process variables across an entire photolithography sequence and receive real-time structural feedback.

2.1. The Digital Twin Paradigm in Semiconductor Fabrication

In advanced semiconductor high-volume manufacturing (HVM), the photolithographic transfer of sub-wavelength design rules is dominated by highly non-linear optical diffraction and complex photoacid diffusion kinetics. Traditionally, predicting wafer-level topography requires Rigorous Coupled-Wave Analysis (RCWA) and iterative finite element methods. While highly

accurate, these solvers are computationally prohibitive, creating massive bottlenecks in verification cycles. The deployment of a Lithography Digital Twin demonstrates a scalable proof-of-concept to disrupt this workflow. By abstracting the physical fab environment into a virtual replica, students and engineers can experience this whole process both from interactive and process parameters variations.

3. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

3.1 System Resolution & Pattern Capabilities

- Target Resolution Nodes: Sub-100nm critical dimensions (CD), capturing structural limits from 40nm up to 150nm feature sizes.
- Supported Layout Topologies: Dense Vertical Gratings (Nominal 93nm Trench target), Dense Horizontal Gratings, Two-Dimensional Contact Arrays, and Isolated Lines.

3.2 Step-by-Step Module Operating Ranges

- RCA SC1 Clean & HMDS Vapor Prime: Executes an industry-standard physical undercut clean for prime temperature (90°C to 160°C) and duration (20s to 300s) to output the final particle removal metrics and surface contact angle.
- Photoresist Spin Coating: Operates within a 120.0nm to 350.0nm thin-film boundary. Takes user input for spin speed (RPM) and time to output the final spatial thickness map.
- Soft Bake (PAB): Establishes structural stability and drives off residual solvent. Outputs a stabilized uniform matrix array mapped across a 1996.8nm x 1996.8nm spatial area.
- Optical Exposure (Stage 1 ML Core): Simulates 193.0nm Argon Fluoride DUV diffraction. Accepts continuous user sweeps for Numerical Aperture (0.50 to 1.35), Defocus (-45.0nm to +45.0nm), and Exposure Dose (22.0 to 38.0 mJ/cm²).
- Post-Exposure Bake (Stage 2 ML Core): Models localized photoacid generation and thermal catalytic deprotection to output a 2D Chemical Protection Profile mapping the remaining polymer bonds.
- Volumetric Chemical Development: Executes fluidic etching erosion using Mack Model Dissolution Primitives (Rmax = 120.0nm/s, Rmin = 0.05nm/s) over a user-defined wash duration of 10.0s to 80.0s.

3.3 Scene Hierarchy & Organization (LabScene)

The primary Unity scene, ‘LabScene’, is structured into distinct hierarchy groups to manage performance and logical separation:

- Environment & Rooms: Static mesh geometry defining the walls, floors, partitions, doors, and yellow lighting fixtures for the two cleanroom bays.
- Workstations & Process Equipment: Interactive GameObject hierarchies representing the Spin Coater, Hotplates, Developer Station, Wet Benches, Fume Hoods, and the EVG Mask Aligner machine.
- Interactables & Wafer Carriers: Objects including wafer substrate, wafer carrier, photomask, and placement docking triggers are all interactable.
- UI Canvas & Monitors: World-space control panels mounted on workstation geometry, screen-space HUD overlays, step instruction banners, and metrology graphing displays.
- Logic & Manager Empties: Dedicated root GameObjects holding centralized coordination scripts, including WorkflowManager, LabProcessManager, and LithoInferenceEngine.

3.4 Supporting Assets & Resources

The project includes curated secondary assets to support simulation realism without introducing unnecessary overhead:

- In-Game Documentation: Text-based operational manuals and reference tables accessible from UI panels implemented using UXML/USS framework
- UI & Photomask Textures: Industry-standard photomask layout textures (contact holes, line-space arrays) and cleanroom signage.
- Material Shaders: Custom HLSL/ShaderGraph shaders for photoresist film rendering, silicon reflectivity, and yellow-light surface diffusion.

4. CLEANROOM ENVIRONMENT & SPATIAL DESIGN

The virtual cleanroom is divided into two specialized operational rooms, both illuminated exclusively by yellow photolithography room lighting. In physical semiconductor fabrication, yellow lighting is strictly enforced because standard white light contains ultraviolet (UV) and blue wavelengths that prematurely expose and ruin photoresist chemical films.

4.1 Room Layout & Equipment Partitioning

- Room 1 (General Processing Bay): Contains the majority of the laboratory workstations. This includes the Spin Coater bench, Soft Bake and Post-Exposure Bake (PEB) Hotplates, the Developer Station, and wafer carrier.
- Room 2 (Mask Aligner Bay): A dedicated, isolated room containing only the industrial EVG Mask Aligner workstation and its associated photomask storage tray. Separating the optical

alignment system into a dedicated space mirrors industrial fab layouts and allows users to focus entirely on complex optical alignment tasks. In addition, references online of real cleanrooms support the partition of these items.

4.2 Spatial Navigation & Interaction

The application utilizes first-person spatial navigation to simulate the physical presence of working inside a cleanroom. Users navigate the environment using standard movement controls and interact with equipment by walking up to workstations and engaging via line-of-sight ray casting. Wafer handling is performed by picking up silicon wafers from carriers, transporting them across the room, and placing them onto designated chucks or surfaces and interacting with the UI of different systems.

5. WORKSTATION DESIGN & MECHANICAL ENGINEERING

Each workstation was designed to balance mechanical accuracy with responsive user interaction.

5.1 HMDS Vapor Prime oven

Features interactive UI sliders for setting the internal bake temperature and prime duration. Visual feedback is presented via a dynamic control panel that continuously calculates and displays the predicted surface contact angle based on the underlying empirical adhesion model. The workstation includes kinematic animations for the oven door to simulate the sealed vacuum chamber required for the combined dehydration and vapor priming process.

5.2 Spin Coating & Baking Workstations

- Spin Coater: Features interactive control knobs for setting rotational velocity (RPM) and dispense duration. Visual feedback is handled via shader logic directly on top of the wafer mesh, representing liquid photoresist expanding and thinning across the surface during rotation. To maintain rendering performance and simulation accuracy, centrifugal waste splatter particle systems were omitted.
- Hotplate Station: Dedicated hotplate for soft baking and post-exposure baking feature temperature controllers and timers. During the baking process, the photoresist shader undergoes a clear color change, to visually indicate temperature-induced solvent evaporation.



Figure 1: Lab bench consisting of HMDS oven, Hotplate, Spincoater and Wafer carrier. RVE display on the TV above.



Figure 2: HMDS prime oven

5.3 Industrial Mask Aligner Modeling & Rigging

Because accurate, open-source 3D models of industrial mask aligners were unavailable, an EVG-inspired alignment system was modelled from scratch in Blender using vendor documentation, and reference videos.

- Mechanical Rigging: The model includes internal and external structural details such as pneumatic tubing, electrical wiring looms, optical microscope tracks, vacuum chuck stages, and motorized photomask trays.
- Kinematic Animations: When imported into Unity, the rig was animated using AlignerAnimationController.cs to execute precise mechanical sequences: automated tray

ejection for mask loading, tray insertion, vacuum chuck wafer clamping, mechanical leveling, optical microscope engagement, and UV exposure lamp activation.

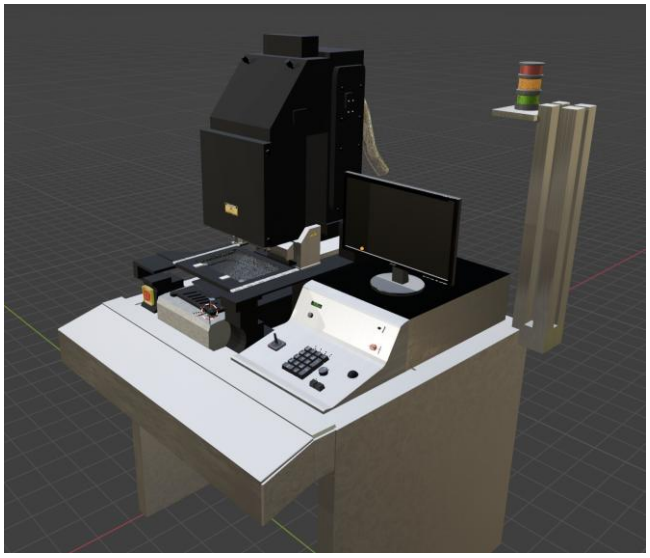


Figure 3: EVG Mask Aligner model in Blender

5.4 Wet Bench & Developer Station

The station comprised of a hotplate and a beaker for SC1 cleaning, sink, two beakers, three bottles and an interactable UI controller. The labelled developer solution (TMAH) bottle and the Deionised (DI) water is present next to the beakers. A bottle containing SC1 solution is placed near to the hotplate. Cleaning using the SC1 Solution is the first step performed here and the developer solution washing at the end. The user is prompted to pour the liquid in along with the wafer, and the controls allow them to change the time the wafer is immersed and swerved.

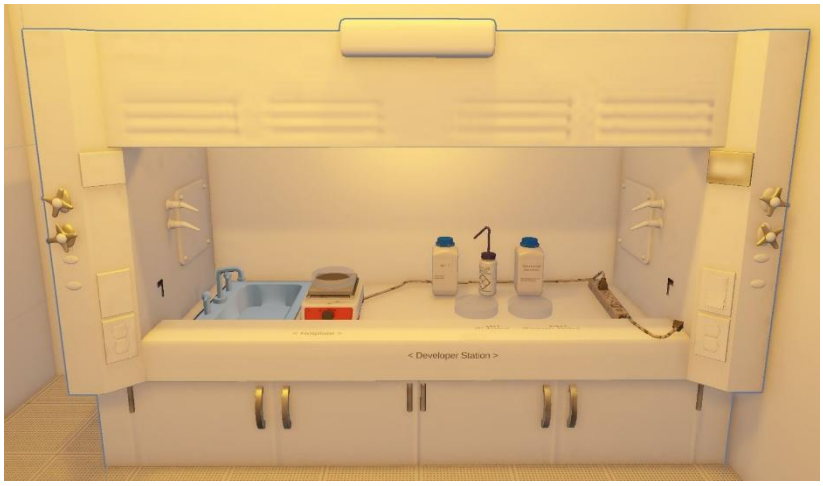


Figure 4: Developer station

6. Pre-Lithography Cleaning

The wafer arriving at the cleaning station has already undergone earlier fabrication step called oxide deposition (SiO2). These steps leave the wafer surface covered in residue: organic contamination from handling, particulate matter from thin molecular films from process chemicals, and leftover deposition byproducts sitting on the oxide surface.

6.1 Stage 1: RCA SC-1 Wet Clean

SC1 stands for Standard Clean 1, the first step in the RCA cleaning sequence developed at RCA Laboratories. The bath is a mixture of ammonium hydroxide (NH_4OH), hydrogen peroxide (H_2O_2), and deionized water in a 1:1:5 volume ratio, also called APM (Ammonia Peroxide Mixture). This typically removes organic particulates, surface-adsorbed molecular films etc.

SC1 does not dissolve organic contamination chemically the way a strong acid would. Instead, it exploits a physical mechanism called undercut lift-off. The NH_4OH very slowly etches the native silicon oxide (SiO_2) surface at approximately 4 Å per minute. As this etch front moves forward, it travels laterally underneath the base of any contamination particle or organic patch sitting at the oxide surface, progressively cutting away its contact area. Once the etch has undercut enough of the particle's footprint, adhesion fails and the particle simply detaches and disperses into the bath liquid. While NH_4OH etches the oxide, H_2O_2 simultaneously re-oxidizes any bare silicon exposed by that etch, maintaining a thin self-replenishing chemical oxide layer of approximately 6–10 Å throughout the entire dip.

The widely cited SiO_2 etch rate at 80 °C for the standard 5:1:1 recipe is 4.0 Å/min. Particle removal saturates once the accumulated etch depth crosses a critical undercut threshold, d_{ref}, of 40 Å.

6.1.1 Chemical Reactions & Etch rate relations

Reaction 1 — NH_4OH etches the oxide (undercuts and removes particles):

$$SiO_2 + 2OH^- \rightarrow SiO_3^{2-} + H_2O$$

Reaction 2 — H_2O_2 re-oxidises bare silicon and water converts it to Si–OH (the hydrophilic surface that HMDS later reacts with):

$$Si + 2H_2O_2 \rightarrow SiO_2 \xrightarrow{H_2O} Si\text{–}OH$$

The first equation is why particles come off. The second equation is why the surface ends up as Si–OH at $\theta \approx 5$–$6°$, which directly sets up the HMDS step.

Equation 1 — Etch depth accumulation:

$$d(t) = r \times t$$

Where r = 4.0 Å/min (Watanabe 1983), t = dip time in minutes.

Equation 2 — Removal fraction:

$$R(t) = \min\left(1.0,\ \frac{d(t)}{d_{\text{ref}}}\right) \times 100\%$$

Where d_ref = 40 Å (critical undercut depth for complete lift-off).

6.2 Stage 2: HMDS Vapor prime

Following the SC1 clean, the SiO_2 surface is terminated with silicon hydroxyl (Si–OH) groups, which renders the surface hydrophilic and prevents proper photoresist adhesion. The wafer is placed into a HMDS Prime Oven.

A vacuum venturi is used to generate the lower pressure in the reaction chamber and the bottom of the chamber is a hotplate that elevates the temperature of the wafer. While this process may not eliminate all of the water on the surface of the wafer, particularly the chemically bound water, it does remove a large percentage. The wafer is now ready for the HMDS reaction portion of the process.

During the HMDS (hexamethyldisilazane) reaction, the process chamber is flooded with HMDS vapor using nitrogen as a carrier gas. HMDS vapor reacts with these surface hydroxyl groups, replacing them with trimethylsilyl groups and releasing ammonia.

$$2\ (CH_3)_3Si\text{–}NH\text{–}Si(CH_3)_3 + Si\text{–}OH \rightarrow Si\text{–}O\text{–}Si(CH_3)_3 + NH_3\uparrow$$

This chemical conversion transforms the hydrophilic surface into a hydrophobic one, raising the water contact angle (θ) and providing the essential adhesion needed for resist spin-coating.

6.2.1 HMDS adhesion model

User Inputs: Temperature (°C), prime time (sec).
Outputs: Contact angle (θ)

To accurately model the contact angle output for the Digital Twin, an empirical saturating exponential model was fitted to historical Design of Experiments (DOE) data as per Fig. 5. The minimum contact angle for acceptable photoresist adhesion (the adhesion gate) is $\theta \geq 65°$. The model calculates the final angle through three phases:

- **Flash Adhesion:** Models the rapid initial reaction starting from the hydrophilic SiO_2 surface.
- **Saturation Limit:** Calculates the maximum possible angle bounded by the user's selected oven temperature.

- **Exponential Decay:** Simulates the time-dependent saturation over the bake duration using an Arrhenius time constant.

Effect = Temperature (°C)	Effect = Prime Time (sec.)	Response = Contact Angle (°)
90	20	61
90	60	64
90	100	68
90	140	72
120	20	63
120	60	66
120	100	69
120	140	73
150	300	77

Figure 5: HMDS characterization DOE parameters

6.3. Unity Implementation and Outputs

SC1 Wet Bench (SC1Controller.cs): Operates on the fixed 80 °C standard recipe. The UI displays real-time undercut etch depth and removal percentages. A time multiplier is utilized to allow players to simulate the 10-minute chemical dip in a fraction of real-world time.

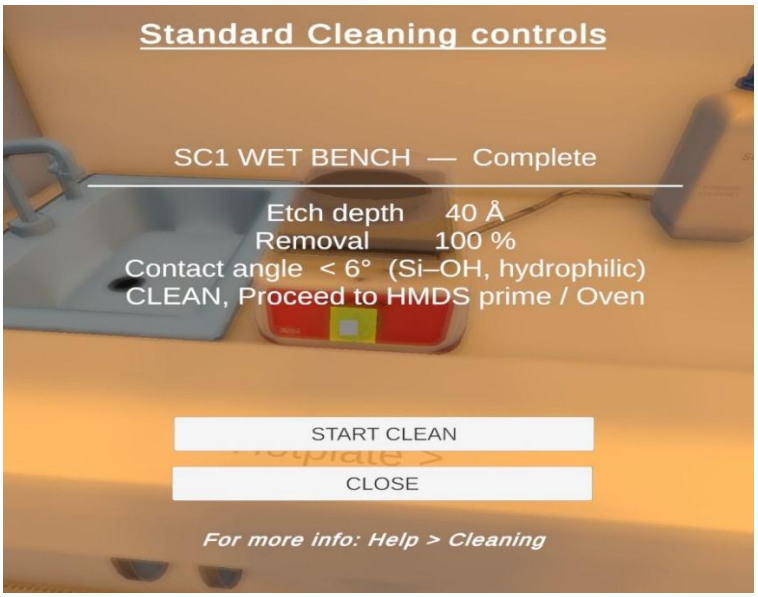


Figure 6: SC1 clean output

Dehydration & HMDS Prime Oven (HMDSController.cs): Allows the user to adjust the combined hotplate Temperature (90 °C–160 °C) and prime Time (20s–300s) via UI sliders. The

empirical model continuously predicts the resulting contact angle. If the user fails to meet the 65° adhesion gate, the system flags a failure and prompts them to recalibrate their parameters, ensuring surface is sufficiently hydrophobic for photoresist adhesion.

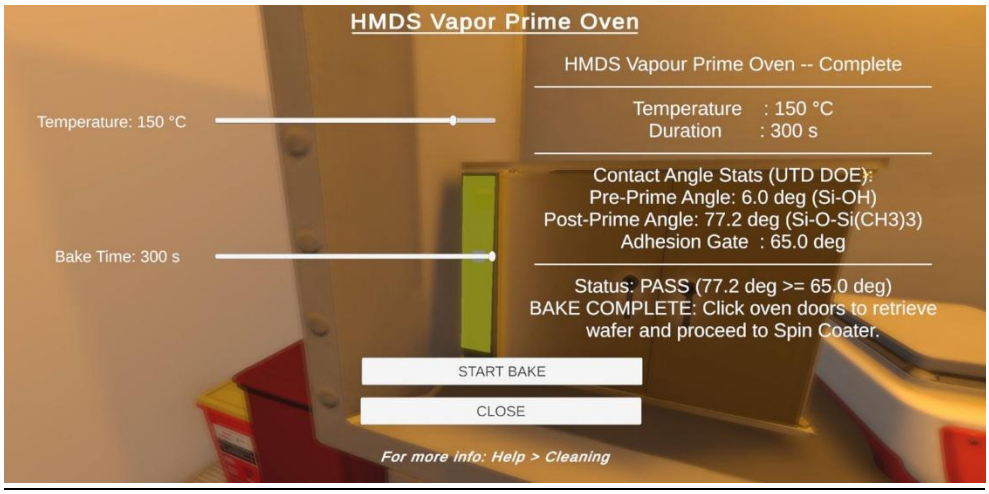


Figure 7: HMDS Vapor prime outputs

7. Process 1: Spin Coating of Photoresist

7.1. Introduction and Concepts of Spin Coating

This section describes spin coating not as a full fabrication-grade physics solver, but as a reduced-order empirical model intended for real-time inference and visualization within the digital twin. Spin coating is one of the most widely used thin-film deposition techniques in semiconductor fabrication, and it is the standard method for applying photoresist layers before lithography. A liquid photoresist is first dispensed onto the wafer surface, and the wafer is then rotated at high angular velocity. The resulting centrifugal force spreads the liquid radially outward, and the film gradually thins until it stabilizes into a uniform coating. For this project, which targets a DUV 193 nm lithography process, the model is built around a target photoresist thickness range of 120 nm to 350 nm, since this is the practical window the digital twin is expected to reproduce.

7.1.1 Process Dependencies

The final photoresist film thickness is governed by several coupled process parameters rather than by any single variable. Spin speed (RPM) and spin time set the mechanical spreading behavior, while photoresist viscosity, fluid density, and surface tension determine how strongly the liquid resists radial flow. Solvent evaporation rate and the acceleration profile used during spin-up influence how quickly the film transitions from a liquid to a semi-solid state, and ambient conditions such as wafer temperature and humidity introduce additional variation that is harder to control directly.

7.1.2 Physical Stages of the Process

Physically, the spin coating process can be divided into five broad stages that occur in sequence once the resist is dispensed:

- Dispense stage: Photoresist is deposited onto the wafer surface.
- Spin-up stage: The wafer accelerates and the fluid begins to spread radially.
- Hydrodynamic thinning: Centrifugal force dominates and the film thickness reduces rapidly.
- Evaporation-dominated stage: Solvent evaporates and the remaining film's viscosity increases.
- Final stabilization: The film reaches a near-steady thickness.

7.2. Evolution of Spin Coating Physics Models

The foundational Emslie model (1958) established the hydrodynamic basis for spin coating by linking film thinning to centrifugal force and viscous flow under idealized, non-evaporative conditions. However, because it ignores solvent evaporation and non-Newtonian fluid characteristics, the original analytical model fails to accurately predict the final stabilized film thickness required for modern semiconductor manufacturing. To bridge this gap, later researchers like Meyerhofer (1978) incorporated evaporation rates and viscosity evolution into the thinning equations. Ultimately, because real resist behavior is highly complex, modern fabs bypass these closed-form physics models entirely in favor of empirical, measurement-derived spin curves for accurate process prediction.

7.3. Spin Curves and the Empirical Model Used

Modern fabs generally do not solve the coupled flow and evaporation equations for every recipe. Instead, they rely on empirical spin curves and direct process characterization: resist vendors supply thickness-versus-RPM curves for standard recipes, and engineers validate or recalibrate these experimentally using ellipsometry, reflectometry, or profilometry. The typical workflow is to choose a resist, collect spin curve data from a datasheet or experiment, and fit an empirical model to extract the required constants.

This project uses the same standard power-law form, $h = K \cdot RPM^{-n}$, in place of the full Emslie relation, since the original does not include evaporation and does not converge to a realistic final thickness, where h is the final photoresist thickness, n is the spin exponent (commonly ≈ 0.5, inherited from the hydrodynamic theory). K is an empirical fitting constant that absorbs everything the simplified equation does not model explicitly, such as resist chemistry, viscosity

and solvent concentration. K was estimated here using a representative DUV target operating point of about 200 nm at 3000 RPM: with n = 0.5, which yields K ≈ 11000 for this resist window. This is a starting calibration value and can be refit once real datasheet or experimental spin-curve data is available.

7.4. Machine Learning Surrogate Model (Input & Output Correlation)

- **Model Architecture:** Multi-Layer Perceptron (MLP) Regression Surrogate.
- **Model Inputs:** Spin Speed (RPM), Spin Time (Seconds)
- **Model Outputs:** Final Photoresist Thickness Map (Targeting a DUV window of 120nm to 350nm).
- **Correlation:** The MLP learns the non-linear relationship between the centrifugal forces (RPM) and the empirical resist properties, outputting a stabilized thickness value that bypasses the complex, time-dependent fluid dynamics and solvent evaporation phases.

7.5. Unity Implementation and outputs

When the user adjusts the UI sliders for rotational velocity and duration, these float values are packaged into a tensor array and passed to the localized MLP surrogate model. The resulting thickness map is then rendered onto the wafer GameObject using a custom HLSL shader, dynamically updating the surface gradient to reflect the final 120 nm to 350 nm film thickness.



Figure 8: Output from the spincoater stage (Speed and Time are inputs)

8. Process 2: Soft-baking (Pre-baking) process

The post-apply bake (PAB) process, alternatively referred to as a softbake or prebake, involves drying the photoresist immediately following the centrifugal spin-coating phase by driving out volatile casting solvents.

The physical elimination of solvent achieves critical process engineering outcomes:

1. It stabilizes film thickness.
2. It optimizes resist adhesion to the underlying substrate matrix.
3. It reduces film tackiness, mitigating susceptibility to particulate defect contamination.

8.1. Governing Textbook Equations

8.1.1 Thermal Decomposition Kinetics

When a positive photoresist is subjected to thermal excitation on a hotplate at temperatures exceeding approximately 70°C to 80°C, its light-sensitive component, the photoactive compound (PAC) undergoes thermal decomposition into a non-photosensitive product ($M \xrightarrow{\Delta} X$).

To quantify the remaining fraction of active sensitizer ($m' = M_0/M_0'$) as a function of processing parameters, we solve the first-order kinetic decay equation:

$$m' = \exp\left[-K_T \cdot (t_b - t_{\text{wup}})\right]$$

Where:

- t_b represents the total softbake time configured by the operator.
- t_{wup} is the system warm-up lag time required for the silicon substrate to reach thermal equilibrium with the hotplate chuck.
- K_T represents the thermal decomposition rate constant

$$K_T = A_r \cdot \exp\left(-\frac{E_a}{R \cdot T}\right)$$

8.1.2 Global Solvent Evaporation and Mass Loss

The physical removal of volatile casting solvents, specifically Propylene Glycol Monomethyl Ether Acetate (PGMEA) from a Poly(p-hydroxystyrene) [PHOST] or acrylic-based chemically amplified resist (CAR) matrix is governed by coupled free-volume diffusion and surface vaporization. Integrated across a thin film, the average remaining solvent concentration ($C(t)$)

over time follows a specialized first-order exponential decay toward a temperature-dependent saturation ceiling (C_{dry}):

$$C(t) = C_{\text{dry}} + (C_0 - C_{\text{dry}}) \cdot \exp[-K_T \cdot t]$$

Where: C_0 is the initial solvent volume fraction directly following spin coating (21% volume fraction), C_{dry} represents the asymptotic zero-solvent minimum limit trapped by the collapsing polymer free volume (2% volume fraction).

8.1.3 Geometric Volume-to-Thickness Transformation Law

As the solvent volume fraction decreases, the polymer matrix undergoes structural shrinkage along the Z-axis. The final post-bake geometric thickness (h_{bake}) is calculated by normalizing the remaining solvent fraction against the initial unbaked boundary conditions:

$$\boldsymbol{h_{\text{bake}}} = h_{\text{dry}} + (h_{\text{spin}} - h_{\text{dry}}) \cdot \left[\frac{C(t) - C_{\text{dry}}}{C_0 - C_{\text{dry}}}\right]$$

Where: h_{spin} is the initial wet thickness as per the spin coater, h_{dry} represents the absolute minimum structural thickness floor of the completely dried solids matrix, calculated via the solids packing fraction: $h_{\text{dry}} = h_{\text{spin}} \cdot [1.0 - (C_0 - C_{\text{dry}})]$

Using this solvent diffusion model, one can predict the impact of pre-bake time and temperature on the final resist thickness and the resulting solvent distribution as a function of depth into the resist at the end of the bake. Such modeling provides for a very interesting result. Any combination of initial resist thickness and initial solvent content that produces a certain final resist thickness always results in essentially the same solvent distribution in the resist at the end of the bake. [11].

8.2. Machine Learning Surrogate Model (Input & Output Correlation)

To map this complex solvent diffusion and geometric shrinkage to a 2D digital twin layout, a Split Neural Network is utilized.

- **Model Architecture:** Split Neural Network.
- **Model Inputs:**
    - Initial Photoresist Spin Thickness (h_{spin}, e.g., 200nm).
    - Hotplate Bake Temperature (T_{user}).
    - Bake Time (t_b).

- **Model Outputs:**
    - 2D Spatial Film Height Map (calculated over a 512×512 coordinate grid).
    - Residual Solvent Volume Fraction.
    - Mean Thickness (h_{avg}).
    - Percent Non-Uniformity Metric (U).
- **Correlation:** The ML model evaluates the theoretical solvent diffusion and thickness transformation equations for the selected temperature and time, outputting a highly accurate spatial grid that maps the final structural shrinkage and solvent distribution of the resist.

8.3. Spatial Uniformity Analysis and ML model outputs

To bridge the 1D neural network output with a high-resolution 2D visual layout for the digital twin, the system runs parallel batch inference over a 512×512 coordinate grid. Thermal non-uniformities across real equipment are accounted for by introducing a radial temperature droop from the hotter center core out to the cooling margins: Once the neural network evaluates the thickness at every grid pixel, the macro-level process quality is verified by computing the **Percent Non-Uniformity (U) Metric** directly from the final spatial thickness map:

$$U = \frac{h_{\text{max}} - h_{\text{min}}}{2 \cdot h_{\text{avg}}} \times 100$$

Where: h_{max} is the maximum thickness detected across the grid pixels (occurring at the cooler outer boundaries), h_{min} is the minimum thickness detected (occurring at the hotter center), h_{avg} is the calculated arithmetic mean height of the entire active wafer surface area.

8.3.1. Outputs for the soft-bake process for various combinations

Figure 9. shows the outputs for the soft-bake stage from the ML model for different combinations of temperature and time which is predicting as per the physical guidelines, which is further integrated with the Unity sliders.

```
======================================================================
DIRECT MAP ONNX RUN: Incoming Flat Spin Base = 200.0 nm
======================================================================

▶ PROFILE READOUT: Low Temp / Short Bake (80.0°C for 30.0s)
  ├─ Center Pixel Film Height:        187.23 nm
  ├─ Center Pixel Residual Solvent:   5.99% Vol
  ├─ Calculated Mean Thickness (Avg): 187.98 nm
  └─ Uniformity Error Metric (U):     0.396%

▶ PROFILE READOUT: Standard Target Window (95.0°C for 40.0s)
  ├─ Center Pixel Film Height:        169.34 nm
  ├─ Center Pixel Residual Solvent:   5.52% Vol
  ├─ Calculated Mean Thickness (Avg): 169.75 nm
  └─ Uniformity Error Metric (U):     0.244%

▶ PROFILE READOUT: Aggressive / Over-Bake (115.0°C for 90.0s)
  ├─ Center Pixel Film Height:        162.59 nm
  ├─ Center Pixel Residual Solvent:   5.15% Vol
  ├─ Calculated Mean Thickness (Avg): 162.59 nm
  └─ Uniformity Error Metric (U):     0.003%
```

Figure 9: ML model output for the soft bake of 200nm initial PR coating

8.4. Unity Implementation and outputs

The post-apply bake sequence is managed by the HotplateController.cs module. As the user initiates the bake timer, the Split Neural Network calculates the 2D spatial film height map and residual solvent volume in the background. To communicate this non-linear evaporation process visually, the wafer's material properties transition through a color shift, confirming the physical stabilization of the photoresist matrix prior to optical exposure.

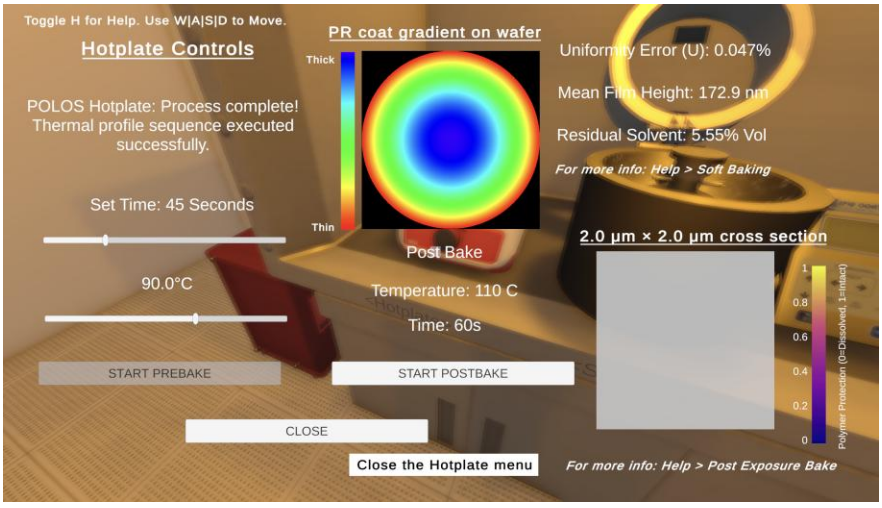


Figure 10: Output from the hotplate after soft baking (Inputs: Time and Temperature)

9. Process 3: Mask generation & Aerial image formation

This digital twin models a **193nm Argon Fluoride (ArF) Immersion Step-and-Scan System** (analogous to industrial ASML Twinscan NXT scanner platforms). DUV light waves stream from the source, passing through illumination optics to establish a uniform partially coherent spatial mode. This light passes through the photomask (object plane), where structural patterns diffract the incoming photons into a wide spectrum of spatial frequency angles. These diffracted fields travel onward to meet the entry pupils of the projection objective lens. This objective assembly lens and mirror combination operating in an immersion fluid medium (ultrapure water, $n = 1.44$). The objective assembly captures the diffracted wave angles, recombining them at the focal plane via inverse Fourier summation to yield the continuous spatial profile known as the **Aerial Image Intensity**.

The physical system constants are strictly bound within the execution script as follows:

- **Exposure Source Wavelength (λ):** 193.0 nm (Deep Ultraviolet Excimer Laser).
- **Total Matrix Coordinate Domain:** 512×512 pixels ($Res \times Res$), 3.9 nm per pixel, defining a total real-world layout canvas footprint of:

$$\text{Field Width} = 512 \times 3.9 \text{ nm} \approx 2.0 \ \mu\text{m}$$

- **Partial Coherence Factor (σ):** 0.85, specifying the ratio of the numerical aperture of the illumination system (condenser) to the numerical aperture of the objective lens.

9.1. Process parameters & Mask datasets generation

- **Exposure Dose:** The total optical energy delivered per unit area to the photoresist film. It controls the population density of generated photo-acids, dictating how deeply and rapidly the chemical boundary changes.
- **Lens Defocus:** The vertical physical displacement of the wafer relative to the ideal focal plane of the projection lens. Defocus introduces phase errors into the light wavefront, blurring the image.
- **Numerical Aperture (**NA**):** The light-gathering power of the projection optics. A higher NA captures higher-angle diffracted light rays to print finer features, but significantly narrows the depth of focus.

Synthetic Mask Dataset Generation

To train and validate the machine learning models across diverse layout configurations, a synthetic dataset of 500 total samples was constructed. The dataset is evenly split into 100 variations per class across five configurations

- Dense Vertical & Horizontal Gratings (Layouts 0 & 1): Symmetrical 7-bar repeating arrays used to evaluate directional, high-frequency line-space diffraction limits and line nesting effects.
- Symmetric Contact Hole Array (Layout 2): An 8×8 dense matrix of 64 square via openings used to evaluate complex multi-directional light interference and corner rounding.
- Isolated Single Vertical Line (Layout 3): A standalone vertical trench representing uncoupled line behavior, typical of transistor gate tracks.
- Isolated Single Contact Via (Layout 4): A single isolated square window used to track baseline uncoupled 2D energy profile transitions.

9.2 Hopkins-Abbe Frequency Domain Mathematics Core

To simulate a partially coherent system quickly, the forward model uses the **Hopkins-Abbe Transmission Cross Coefficient (TCC) Formulation**. This method decomposes the extended light source into a discrete array of mutually incoherent point-source nodes. Each node emits a coherent wave plane that undergoes independent Fourier diffraction routing.

Step 1: Discrete Frequency Space Transformation

A 2D discrete frequency mesh grid (f_x, f_y) is mapped out using the sample pitch resolution ($\Delta x = 3.9$ nm): $\boldsymbol{f}_{\text{radius}} = \sqrt{f_x^2 + f_y^2}$

Step 2: The Complex Pupil Operator & Wavefront Aberrations

The objective lens system acts as a low-pass frequency filter, truncating high-angle diffraction orders that escape past its numerical aperture (NA). The finite physical lens boundary is explicitly tracked via the binary circular **Pupil Function**:

$$P(f_x, f_y) = \begin{cases} 1.0, & f_{\text{radius}} \leq \dfrac{NA}{\lambda} \\ 0.0, & otherwise \end{cases}$$

The wavefront undergoes phase deformation when the lens moves away from the ideal focal plane. This shift is governed by a quadratic optical **Defocus Aberration Function** (W):

$$W(f_x, f_y) = 0.5 \cdot \text{Defocus} \cdot \lambda^2 \cdot f_{\text{radius}}^2$$

Combining both parameters yields the **Complex Pupil Operator** (H_{pupil}), which models the low-pass truncation behavior alongside focus blur phase shifts:

$$\boldsymbol{H}_{\text{pupil}}(\boldsymbol{f_x}, \boldsymbol{f_y}) = P(f_x, f_y) \cdot \exp\left(j \cdot \frac{2\pi}{\lambda} \cdot W(f_x, f_y)\right)$$

Step 3: Concept of the Spatial Frequency Cutoff

The maximum spatial frequency that can propagate through the objective optics is bounded by the lens aperture limit:

$$f_{\text{cutoff}} = \frac{NA}{\lambda}$$

9.3. Partially Coherent Source Discretization & Summation

The extended illumination source is defined by the partial coherence factor (σ), which establishes the maximum illumination spatial frequency boundary ($f_{source_limit} = \sigma \cdot \frac{NA}{\lambda}$). To simulate this extended source, the continuous circular domain is discretized into a uniform 2D grid of mutually incoherent point sources.

For each active point source, its oblique angle of incidence effectively shifts the diffracted wave spectrum in the frequency space. To compute this coherent contribution, the system first transforms the binary mask geometry into the frequency domain via a Forward 2D Fast Fourier Transform (FFT). The resulting frequency matrix is then spatially shifted by an offset directly proportional to the incident angle of the specific point source node.

9.4. Pupil Filtering & Inverse Reconstruction: The shifted diffracted wave fields pass through the objective pupil filter. They are then mapped back to spatial coordinates via an Inverse 2D Fast Fourier Transform ($\mathcal{F}^{-1}$):

$$E_{\text{coherent}}(x, y) = \mathcal{F}^{-1}\{O_{\text{shifted}}(f_x, f_y) \cdot H_{\text{pupil}}(f_x, f_y)\}$$

9.5. Incoherent Square-Law Integration: Because the point sources are mutually incoherent, their resulting electric field amplitudes cannot create interference patterns with one another. Instead, their individual *intensities* (the square of the complex field amplitude) sum up over all valid source entries (N_{nodes}) to construct the total **Aerial Image Intensity (I_{aerial})**:

$$\boldsymbol{I}_{\text{aerial}}(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{N_{\text{nodes}}} \sum_{n=1}^{N_{\text{nodes}}} \left|E_{\text{coherent},n}(x, y)\right|^2$$

9.6. Machine Learning Surrogate Model (Input & Output Correlation)

- **Model Architecture:** Native NCHW Convolutional U-Net.

- **Model Inputs:** A 4-channel spatial tensor (Shape: $4 \times 512 \times 512$) containing: Binary Mask Matrix, Exposure Dose (E), Defocus, Lens Numerical Aperture (NA).
- **Model Outputs:** A 1-channel continuous tensor (Shape: $1 \times 512 \times 512$) representing the Aerial Field Intensity ($0.0 \leq I \leq 1.0$).
- **Correlation & Training:** The U-Net maps the physical scanner parameters and mask geometries directly to the diffracted aerial image. To track high-frequency diffraction waves, the macro-architecture utilizes a wide feature map capacity (scaling from 64 to 256 channels) and LeakyReLU activations to prevent neuron death in areas of optical darkness. The network was optimized using a blended global loss function (80% MSE + 20% L1 Loss) to preserve true optical contrast degradation.

9.7. Unity Implementation and outputs

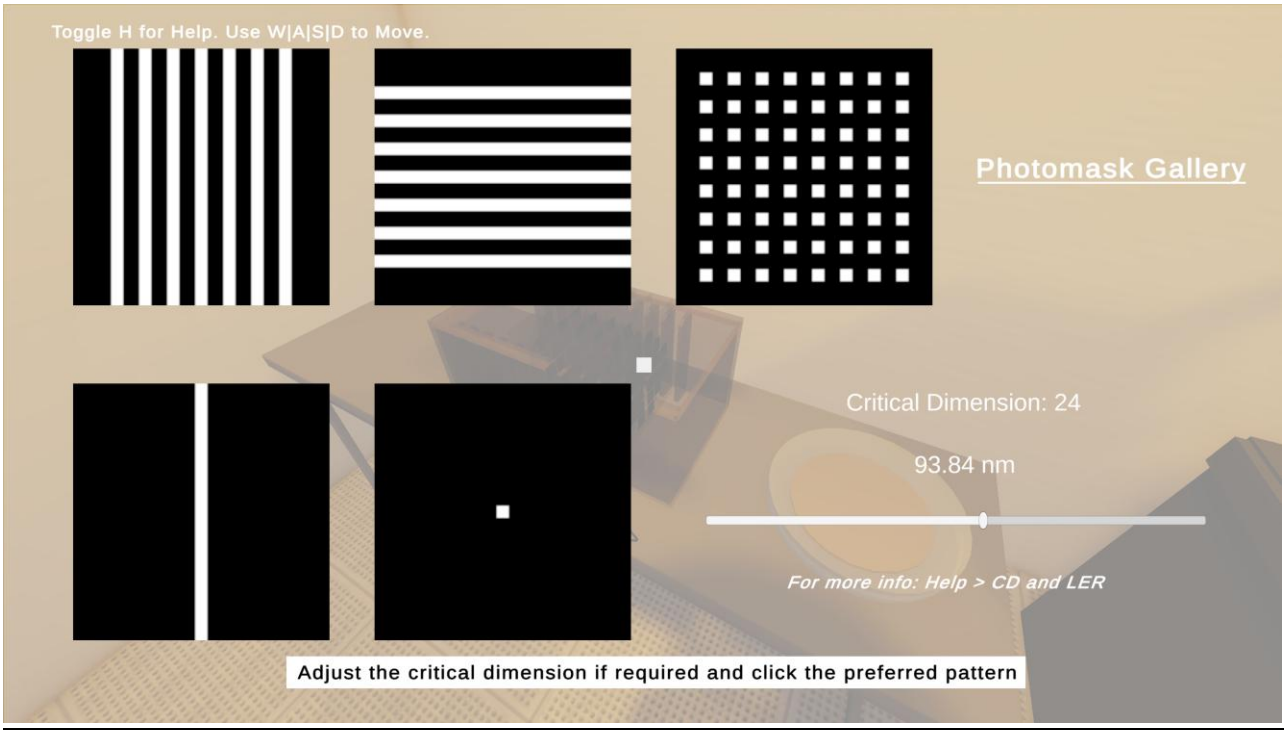


Figure 11: Photomask gallery where user can choose the pattern and adjust Critical Dimension

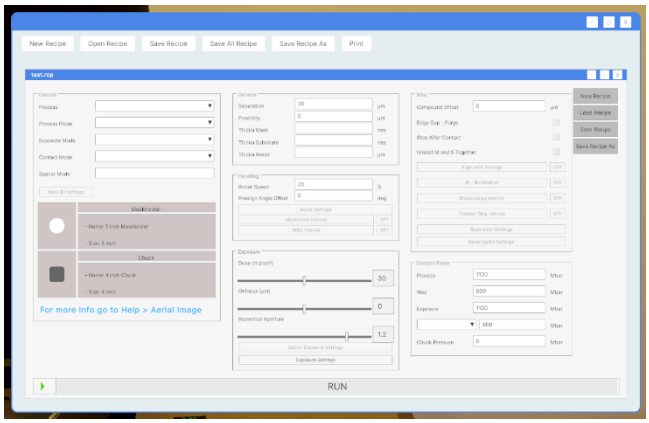


Figure 12: User being able to adjust dose, defocus and NA



Figure 13: Last Stage of Mask Alignment

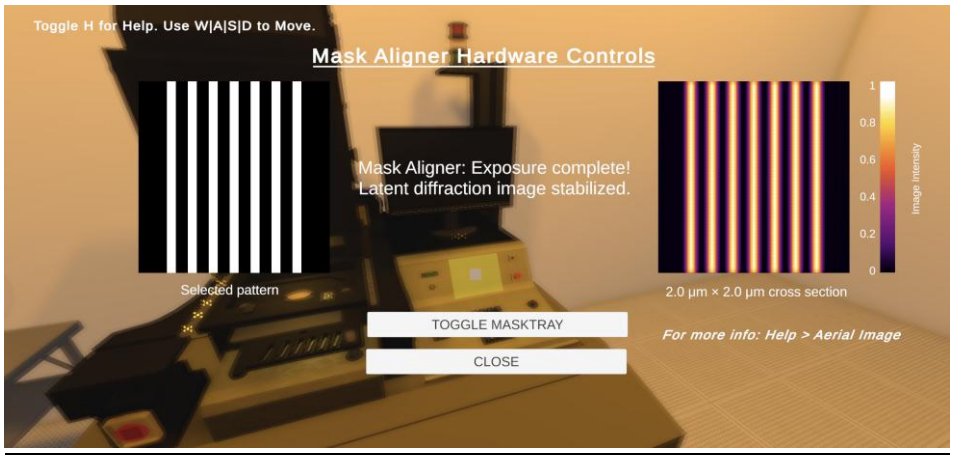


Figure 14: Output from the exposure stage

Translating the Hopkins-Abbe optical physics into real-time gameplay relies on the LithoInferenceEngine.cs script and the Unity Sentis framework. The user's selected photomask, alongside their custom Dose, Defocus, and NA parameters, are compiled into a $4 \times 512 \times 512$ input tensor. The Sentis engine processes this through the ONNX-exported U-Net on the local GPU, instantaneously rendering the resulting Aerial Image Intensity onto the mask aligner's digital interface as shown in Fig. 14.

10. Process 4: Post-Exposure Bake & Development

In modern 193 nm DUV lithography, the exposure step alone does not alter the physical solubility of the photoresist film. The ultraviolet photons lack the energy required to break the

polymer protecting groups directly; instead, they interact exclusively with embedded Photoacid Generator (PAG) molecules.

The primary function of PEB is to use controlled baking temperatures and durations to enhance and drive the chemical reactions initiated during exposure. The thermal energy from the hotplate forces the generated acid molecules to diffuse through the film and act as catalysts for subsequent reactions. This process increases the solubility contrast between the exposed and unexposed areas, rendering the deprotected resin hydrophilic and soluble in an alkaline developer solution, which is crucial for achieving high-resolution patterns.

Emergence of the Final Resist Profile **(M):** At the conclusion of the PEB step, first-order reaction kinetics yield the finalized concentration of the remaining protected polymer matrix fraction (M). The local value of M represents the remaining chemical fraction (ranging between 0.0 and 1.0) that determines the local solubility contrast.

10.1. Phase I: Exposure & Acid Generation (Dill's ABC Law)

When DUV light photons strike the photoresist, they interact with the Photoacid Generator (PAG) compounds embedded in the polymer matrix. This photochemical response is governed by a simplified variation of Dill's A/B/C kinetic parameters. Assuming a negligible change in resist absorbance during exposure (highly applicable to thin-film chemically amplified resins), the local concentration of generated photoacid H_0 immediately following exposure is given by:

$$H_0(x, y) = 1.0 - \exp(-C \cdot \boldsymbol{I}(\boldsymbol{x}, \boldsymbol{y}) \cdot E)$$

Where: C: Dill's activation parameter (*DILL_C = 0.055*), representing the photolysis rate constant (cm^2/mJ), $I(x, y)$: Local normalized aerial image optical intensity (*intensity batch*), E is the absolute exposure dose delivered to the field in mJ/cm^2 (*dose_mJ*).

Note: Following the physical framework derived by Chris A. Mack (*Fundamental Principles of Optical Lithography*, Chapter 6, Eq. 6.6–6.9), assuming localized acid diffusion variations are balanced over short intervals and acid loss is treated as negligible, the acid profile remains locked to its initial exposure footprint ($H(t) = H_0$). Integrating the deprotection kinetic equation yields the definitive textbook relationship for the normalized remaining blocked site fraction (m):

10.2. Phase II: Post-Exposure Bake (PEB) Kinetics

During the Post-Exposure Bake thermal phase, two competing actions occur simultaneously: thermal acid-catalyzed deprotection of the polymer resin matrix and base quencher neutralization.

10.2.1. Coupled Reaction-Diffusion PDEs

The spatial-temporal progression of the photoacid $H(x, y, t)$ and base quencher $Q(x, y, t)$ concentrations over the bake duration is captured by two coupled non-linear partial differential equations (Fick's Second law of diffusion):

$$\frac{\partial H}{\partial t} = D_H \nabla^2 H - K_N H Q$$

$$\frac{\partial Q}{\partial t} = D_Q \nabla^2 Q - K_N H Q$$

Where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is the 2D spatial Laplacian operator. The chemical parameters are defined as:

- D_H: Diffusivity coefficient of the photoacid (*D_H = 3.0* nm^2/s).
- D_Q: Diffusivity coefficient of the base quencher (*D_Q = 1.0* nm^2/s).
- K_N: Recombination velocity constant, governing the acid-base neutralization rate (0.1).

10.2.2. Catalytic Resin Deprotection Mechanics

The photoacid acts strictly as a catalyst. As it diffuses horizontally through the matrix, it systematically cleaves the bulky ester protecting groups attached to the polymer backbone, rendering the resin hydrophilic and soluble in an alkaline developer solution. The concentration of the remaining protected polymer matrix fraction $M(x, y, t)$ changes according to first-order reaction kinetics:

$$\frac{\partial M}{\partial t} = -K_{\text{AMP}} \cdot H(x, y, t) \cdot M(x, y, t)$$

Where K_{AMP} is the amplification kinetic multiplier (*K_AMP = 0.18*), which determines the rate of catalytic deprotection per unit of available acid.

10.3. Numerical Implementation & Solver Stability

This methodology solves non-linear partial differential equations across a discrete pixel grid using an Explicit Forward Euler Finite-Difference Scheme. The process begins by replicating edge states to pad borders, followed by computing spatial Laplacians with a 5-point central difference stencil, evaluating the acid-base neutralization reaction coupling term. Finally, via explicit Euler integration, the resulting concentrations obtained as shown in Fig. 15.

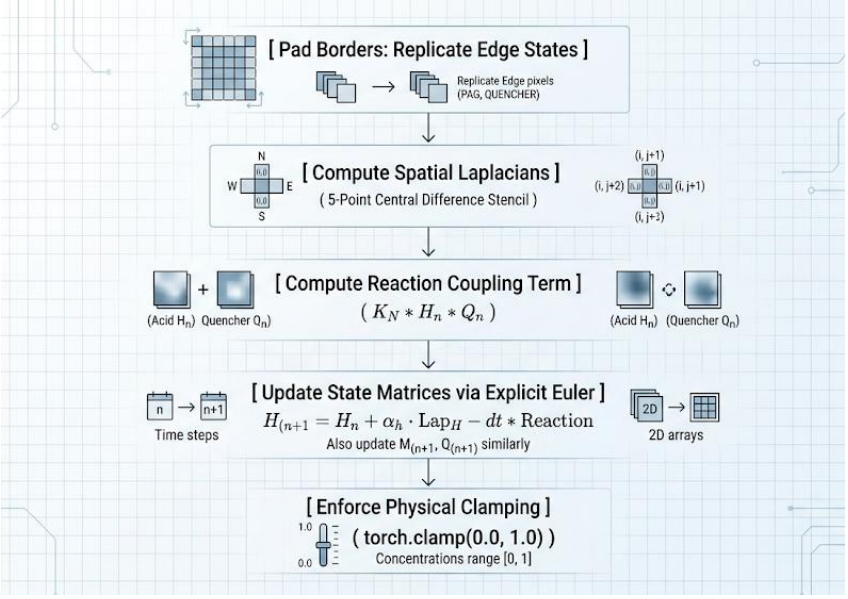


Figure 15: Reaction-diffusion differential solvers

10.4. Deep Physics-Informed U-Net Surrogate Model

While the finite-difference solver is physically accurate, looping through 150 iterations per sample is computationally expensive. This pipeline utilizes a standard Chemical U-Net to predict the final steady-state $M(x, y)$ matrix in a single forward pass as shown in Fig. 16.

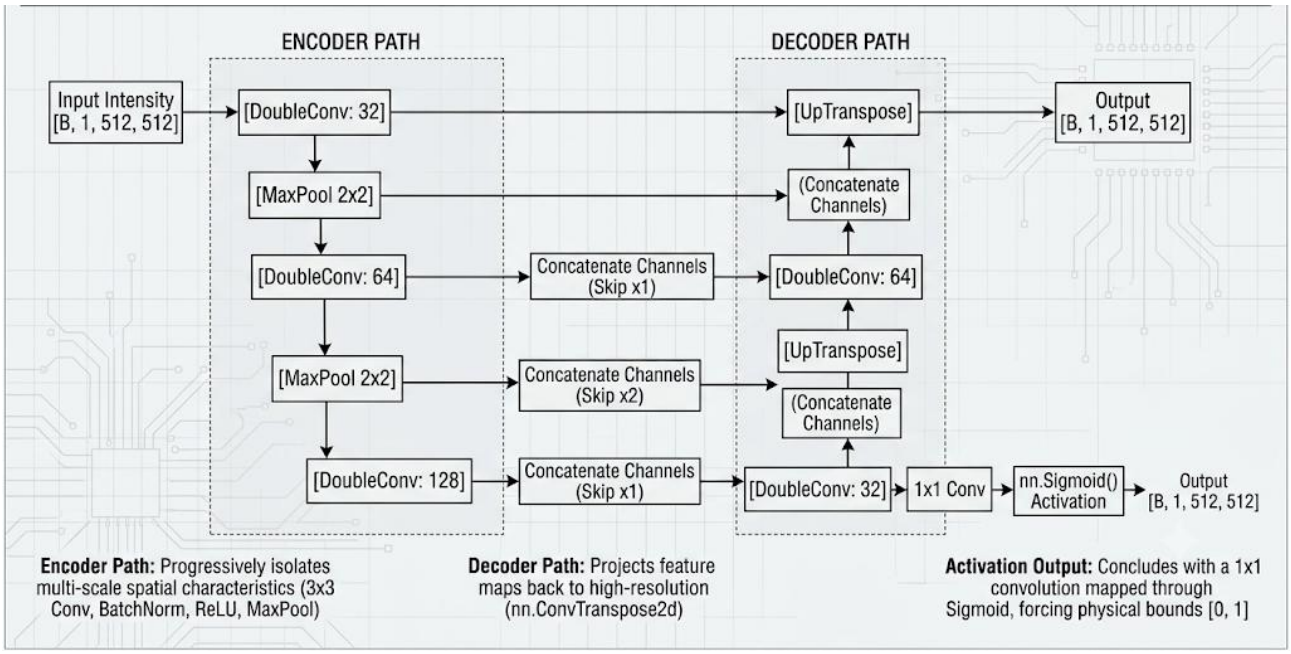


Figure 16: U-net architecture used for ML model

Therefore, a physics-informed U-Net is deployed to predict the steady-state chemical equilibrium in a single forward pass.

- **Model Architecture:** Standard Chemical U-Net (with Double Convolutional blocks).
- **Model Inputs:** A 2-channel spatial tensor (Shape: $2 \times 512 \times 512$) containing:
    - Aerial Field Intensity (propagated directly from the Optical Stage output)
    - Exposure Dose
- **Model Outputs:** A 1-channel continuous tensor (Shape: $1 \times 512 \times 512$) representing the final protected polymer matrix fraction (M). To align with rigorous 3D volumetric physics engines (such as LithoBench-PDE), this 2D output predicts the depth-averaged mean projection of the 3D resist film.
- **Correlation & Training:** The network correlates the initial photoacid generation (driven by the aerial image) to the final deprotected state of the resin. Because chemical deprotection boundaries strictly dictate the final printed Critical Dimension (CD), this network utilizes dual sequential convolutional layers per resolution level to capture complex spatial diffusion. It was fine-tuned using a custom Edge-Weighted MSE Loss function (utilizing a Sobel-style kernel) to heavily penalize chemical prediction errors along the physical layout contours.

10.5. Non-Linear Development Engine (Mack Dissolution Model)

Once the post-bake deprotection profile $M(x, y)$ is generated, it is passed to the **Mack Dissolution Model**. This model relates the local chemical state of the resist to its vertical etching rate in an alkaline developer solution.

- Highly Protected Resin(M=1.0) → Minimal Dissolution Velocity (R_{min} = 0.05 nm/sec)
- Highly De-Protected Resin(M=0.0) → Maximum Dissolution Velocity (R_{max} = 120 nm/sec)

10.5.1. The Kinetic Rate Equations

The dissolution velocity field $R(M)$ in nanometers per second is calculated using the standard four-parameter Mack equation:

$$R(M) = R_{\max} \frac{(a+1)(1-M)^n}{a + (1-M)^n} + R_{\min}$$

Where a is a dimensionless contrast maximization threshold parameter defined by the activation limit constraint M_{th}:

$$a = \frac{n+1}{n-1}(1 - M_{\text{th}})^n$$

10.5.2. Physical Constants and Parameters

Kinetic behaviors are governed by the following configured parameters:

- $R_{\text{max}} = 120.0$ nm/s: The maximum dissolution rate of fully un-protected photoresist resin.
- $R_{\text{min}} = 0.05$ nm/s: The background leakage dissolution rate of completely unexposed photoresist.
- $n = 5$: The developer selectivity contrast exponent. This high value creates a non-linear threshold response that produces near-vertical resist sidewalls.
- $M_{\text{th}} = 0.55$: The threshold activation limit defining the point where the resin transitions from insoluble to highly soluble.

10.5.3. Topography Slicing and Binary Mask Generation

The total thickness of photoresist stripped away during development is calculated by integrating the dissolution rate over the wash time (*WASH_TIME_S = 30 seconds*):

$$\text{removed_nm}(x, y) = R(M(x, y)) \cdot t_{\text{wash}}$$

The final remaining topographical height map is bounded by the substrate floor to prevent negative values:

$$\text{height_map}(x, y) = \max(\text{INITIAL_THICKNESS_NM} - \text{removed_nm}(x, y),\ 0.0)$$

To display this topography on a standard top-down wafer inspection tool, it is converted into a binary matrix step function using a strict substrate clearing threshold:

$$\text{binary_pattern}(x, y) = \begin{cases} 1 & \text{if height_map}(x, y) > 0.0 \quad \text{(Standing Resist Line)} \\ 0 & \text{if height_map}(x, y) = 0.0 \quad \text{(Cleared Silicon Trench)} \end{cases}$$

10.6. Automated Data Analysis & Metrology Architecture

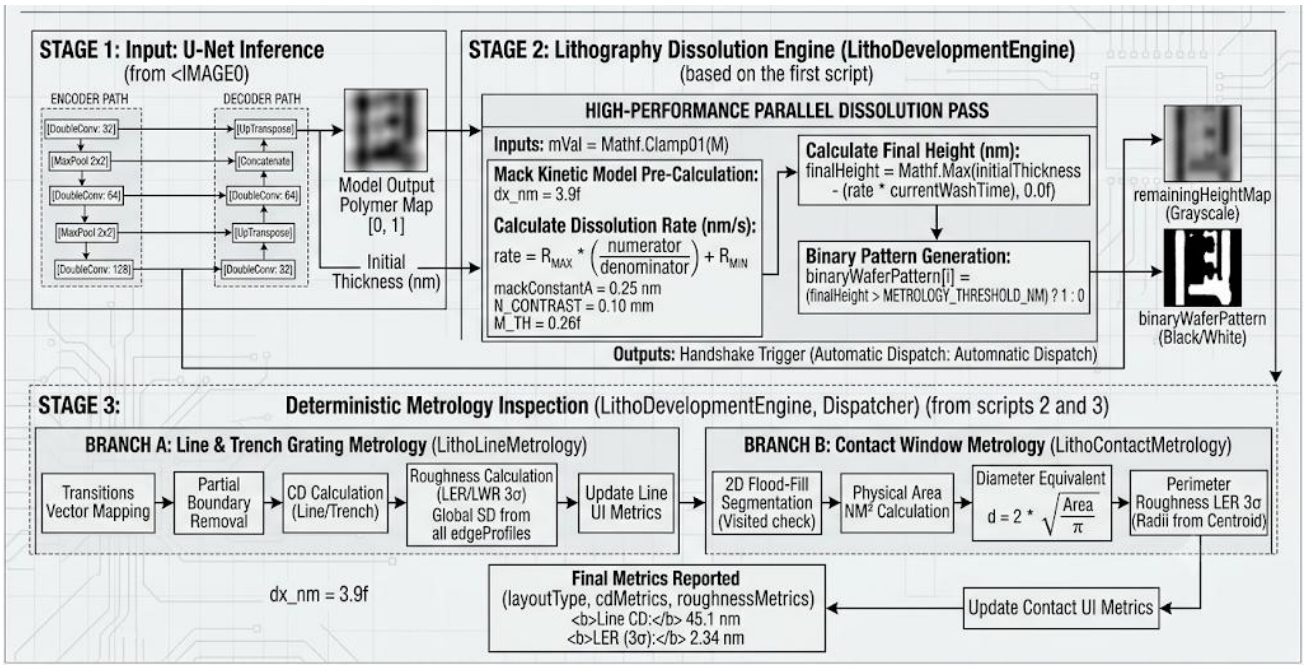


Figure 17: Development metrics (CD and LER)

10.6.1. Critical Dimension (CD) metrology

- Critical Dimension (CD) is the measured width of printed lines or spaces. Physical cleanrooms use expensive, high-voltage CD-SEM tools to track line edges via electron-beam shadow detection. This simulator replaces those rigid tools with pixel-based metrology that scans binary development images.

- CD Extraction: The engine counts continuous pixel runs along a cross-sectional track and multiplies them by the physical grid pitch (3.9 nm per pixel) to output Mean trench CD (average feature width) and Mean line CD (average gap width). And for contact array pattern the CD is calculated using area calculations and diameter as shown in Fig. 17.

10.6.2. Surface Quality Metrology (LER & LWR Derivatives)

- **Line Edge Roughness (LER):** Measures the structural displacement variation of the printed feature edges. It is computed as 3σ (three standard deviations) of the left-most transition coordinate locations across all scanned rows:

$$\text{LER} = 3.0 \cdot \sigma_{\text{edge}} \cdot \Delta x$$

- **Line Width Roughness (LWR):** Measures the variation in total line feature thickness. It is calculated as 3σ of the gathered line width across all rows:

$$\mathrm{LWR} = 3.0 \cdot \sigma_{\mathrm{widths}}$$

10.7. Unity Implementation and outputs

In the backend, the Unity Sentis engine feeds the latent Aerial Image Intensity and Exposure Dose into the Chemical U-Net. In a single forward pass, the model predicts the steady-state protected polymer matrix fraction (M). This 2D chemical profile is immediately rendered on the workstation's HUD, allowing the user to visually verify the catalytic acid diffusion.

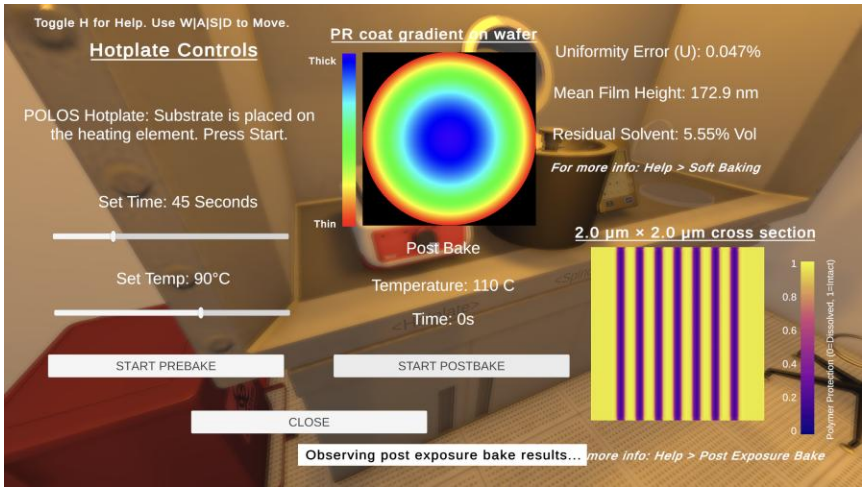


Figure 18: Hotplate output from the post exposure bake stage

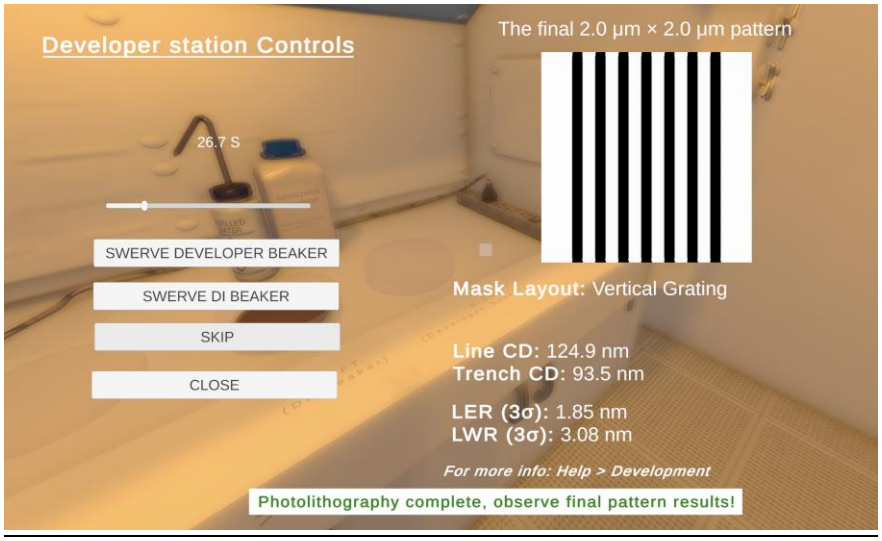


Figure 19: Final pattern with results after Development stage

As the user immerses the wafer and sets the wash duration via the UI, the script applies the non-linear Mack dissolution kinetic equations across the 512×512 grid. Once the development timer

finishes, the engine generates the final binary topography. Simultaneously, the metrology scripts scan the layout to extract the Critical Dimension (CD) and surface roughness.

10.8. Representative Volume Element (RVE) Visualizer

To connect macroscopic workstation controls with sub-micron chemical phenomena, we implemented an RVE visualizer. When operating process benches, users can open the RVE display to observe a real-time cross-sectional simulation of the photoresist film. Driven by Sentis tensor readbacks and custom shaders, the RVE visualizer displays:

- Photoactive Compound Destruction: The degradation of inhibitors in positive resists during UV exposure.
- Acid Diffusion: The thermal migration of generated acids during post-exposure baking to smooth out standing wave artifacts.
- Polymer Dissolution: The selective removal of exposed polymer chains during alkaline development.

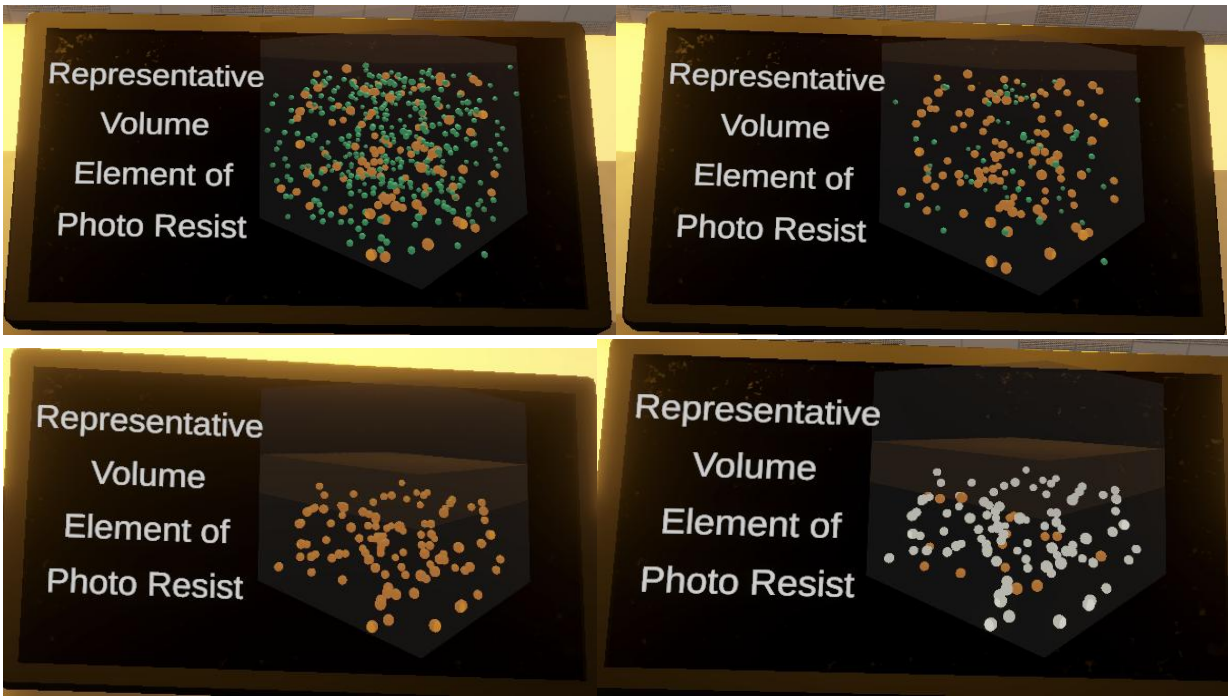


Figure 20: RVE views during dispensing PR, spincoating, soft baking, and exposure

11. END-TO-END VALIDATION & PARAMETER SWEEPS

To test that the model works as expected, we ran the full pipeline, from the optical stage through to the chemical development stage, using different mask designs and process settings. This

validates that the model produces reliable results without the need for the original complex physics simulations.

11.1. System Dashboards

The pipeline successfully turns light intensity (optical stage) into a chemical pattern (PEB stage), and finally into a physical wafer feature (development stage). The dashboards below show this process in action for the mask layouts considered (These plots represent the optimal process parameters yielding ideal topographical outputs).

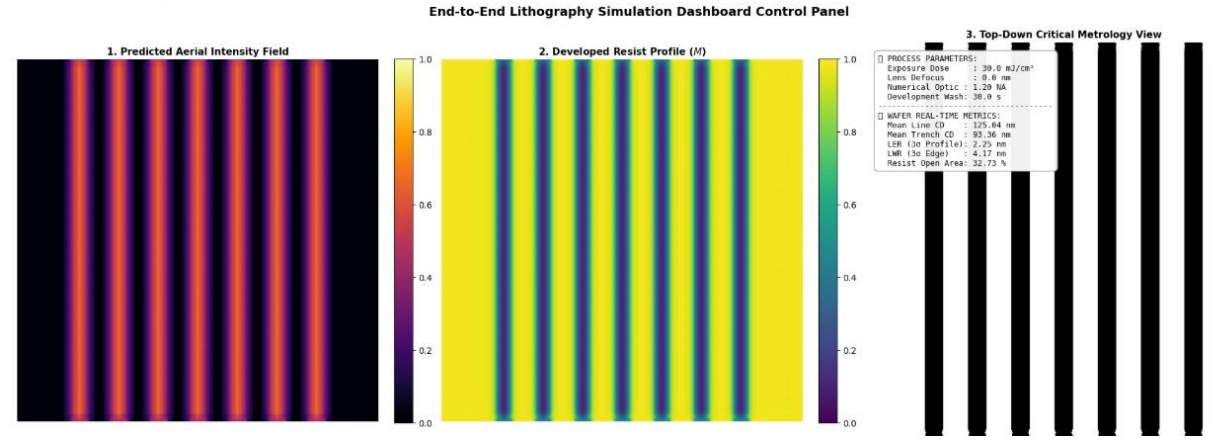


Figure 21: Vertical gratings mask lithography

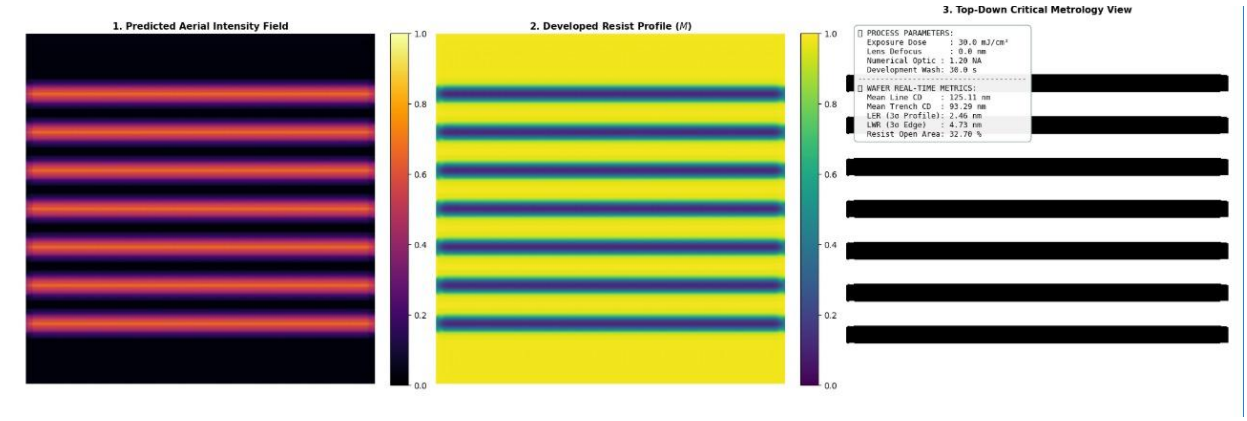


Figure 22: Horizontal gratings mask lithography

11.2. Checking the physics (Validation sweeps)

To make sure our neural network follows real physics rules, we ran "sweeps" where we slowly changed the light dose and development time. We then used Python's matplotlib library to plot these results after the model finished training.

CD vs. Development Time: As we increase the time the wafer spends in the developer fluid, the model correctly computes deeper and wider features. These trends show that the model captures the non-linear way that developer fluid etches the resist as shown in Fig. 23.

CD vs. Exposure Dose: As we increase the light dose, the model correctly predicts a wider footprint for the light-sensitive chemicals. This leads to a wider trench Critical Dimension (CD), which matches the physics described by Dill's laws of photolithography as shown in Fig. 24.



Figure 23: CD versus development wash time

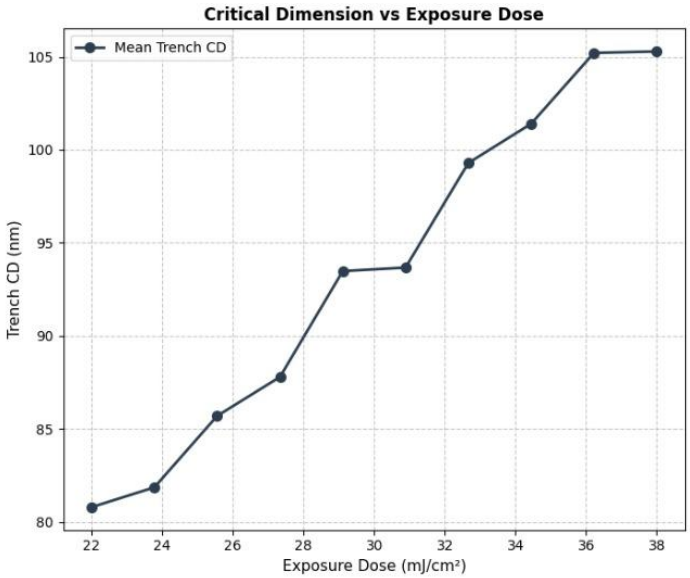


Figure 24: CD versus exposure Dose

12. External Benchmark Validation & Model Limitations

Our machine learning models were trained entirely on a synthetic dataset that we generated ourselves. To verify that our digital twin can handle data it has never seen before, we performed a first-pass validation check against an independent, open-source dataset called **AISDL-SNU/LithoBench-PDE**. The model successfully processed the external layouts. The test successfully validated the model's spatial accuracy while isolating a specific geometric bias:

- **Optical Stage (Aerial Image):** The model showed strong positional correlation (0.912) with the external ground truth; predicted light intensity features aligned closely with the true positions. Structural similarity (SSIM) was more moderate at 0.545, indicating the fine spatial detail differs more than raw position does; we attribute this gap primarily to a physics-model mismatch rather than a positional error.

- **Chemical Stage (Polymer Concentration):** Comparing our predicted polymer map directly against the benchmark's ground truth polymer field revealed a clear qualitative difference in character, not just magnitude. The benchmark's true polymer distribution is broadly graded, with meaningful density spread across the full concentration range. Our predicted output, by contrast, is heavily concentrated near full conversion, behaving close to a binary map rather than a smooth chemical gradient. This represents a fundamental difference in output distribution, rather than a simple numerical offset.

- **Root Cause:** Tracing this back through our own pipeline, we found the cause upstream of the neural network: the physics-based simulator used to generate the Chemical U-Net's training targets uses a cumulative reaction-extent constant that drives nearly any realistic exposure level to near-complete chemical conversion. This leaves little to no representation of intermediate, partially developed states in the training data itself. The network correctly learned this training signal; the signal itself needs correcting at the source.

Note: We do not report downstream thickness or CD accuracy against this benchmark, as LithoBench-PDE provides ground truth at the polymer-concentration level, not a directly measured final thickness or CD value. Any thickness figure would be a derived quantity from applying our own Mack equations to both predicted and true polymer maps, informative for internal diagnosis, but not an independently benchmarked result, so we've kept this section to what was directly measured against the external ground truth.

12.1. Current Limitations and Identified Fixes

- **Training-Data Saturation in the Chemical Stage (identified, fix planned).** The reaction-extent constant governing our training-data generator saturates chemical conversion too aggressively across most of the practical exposure range, leaving intermediate

concentration states underrepresented in training data. Correcting this data generation constraint is the highest priority for improving future chemical-stage accuracy.

- **Lack of Stochastic Noise:** Real-world photolithography at sub-100nm scales is affected by random photon shot noise and stochastic chemical diffusion effects. Our model uses deterministic, continuous math (PDEs) and cannot represent true stochastic line-edge roughness (LER); any roughness visible in our simulation is a digital artifact of the 512×512-pixel grid rather than physical noise. Planned future work includes adding photon shot noise, spatially correlated PAG/quencher non-uniformity, and post-hoc LER to both the optical and chemical stages.

- **Scalar vs. Vector Optical Physics:** Our optical model uses a simplified scalar (Hopkins-Abbe) approximation for light diffraction. The LithoBench-PDE benchmark's ground truth field structure suggests it likely uses a more rigorous vectorial model that accounts for light polarization.

- **Missing Baseline Parameters:** Because LithoBench-PDE does not expose its true process parameters (NA, defocus), some portion of our optical-stage error may reflect testing under incorrect assumed conditions rather than genuine model error.

- **Simplified Dissolution Trajectory:** The development step applies a direct 1D vertical fluidic etching approximation mapped across a 2D grid. True 3D lateral development trajectories, where solvent penetrates sideways into the resist bulk are simplified, slightly altering the exact curvature of the resist footing.

13. USER EXPERIENCE (UX), WORKFLOW & ACCESSIBILITY

13.1 Workflow Manager & Step State Machine

To guide students through the multi-stage cleanroom procedure without confusion, we implemented a centralized ‘WorkflowManager.cs’ script. This state machine tracks user progress across a structured 60-step recipe governing the entire photolithography sequence, ensuring that prerequisites are completed in the correct physical order.

13.2 Contextual Spatial Highlighting

To prevent UI clutter and visual overload, we implemented contextual spatial highlighting via PlacementPulseIndicator.cs. Interactive wafer carriers, machine buttons, and placement docks emit a soft, pulsating visual highlight only when they are the target of the active procedural step. Once the user completes the required action, the highlight turns off immediately.

13.3 Cross-Platform Accessibility & Touch Controls

The application includes responsive input systems designed for desktop, tablet, and mobile browsers. When accessed on touchscreen devices, MouseLook.cs and associated UI managers automatically activate a virtual floating joystick for locomotion and touch-look delta smoothing for camera rotation, allowing smooth navigation without obstructing control panel buttons.

14. WEB DEPLOYMENT ENGINEERING & PERFORMANCE OPTIMIZATION

Deploying a 3D simulation with live 512x512 neural network inference to Unity WebGL presented specific memory and hardware compatibility challenges, particularly for users running browsers on integrated graphics hardware.

14.1 WebAssembly Memory Architecture

During initial scene loading, WebAssembly heap allocations can cause spikes in browser RAM usage. To prevent out-of-memory browser tab crashes, we optimized Unity's WebGL memory configuration:

- Initial Memory Allocation: Tuned to a lightweight baseline of 256–384 MB.
- Maximum Memory Growth Cap: Enforced a strict ceiling of 1024–1536 MB with geometric growth stepping (10–15%). This prevents excessive RAM reservation and forces aggressive browser garbage collection of temporary inference tensors.

14.2 Automated Compute Fallback Architecture

To resolve shader compilation timeouts and fatal WebGL context crashes [WEBGL_lose_context] on older integrated graphics chips, we implemented an automated compute fallback system across all inference controllers (SpinController.cs, HotplateController.cs, and LithoInferenceEngine.cs):

- Startup Hardware Detection: In the Start() method, the application evaluates device capabilities via SystemInfo.graphicsShaderLevel and available video memory. If low-end integrated graphics (shader level < 35 or VRAM < 512 MB) are detected, the engine bypasses GPU shader compilation entirely and initializes Sentis workers using BackendType.CPU (WebAssembly SIMD multi-threaded processing).
- Runtime Try-Catch Failover: When running on standard GPU pixel backends (BackendType.GPUPixel), all inference scheduling and tensor readbacks are wrapped in try-catch error handlers. If a GPU driver times out or fails during a heavy mask alignment execution, the system intercepts the exception instantly, disposes the broken GPU worker,

and re-executes the calculation on the CPU SIMD backend without dropping the graphics context or interrupting visual rendering.

15. CONCLUSION & FUTURE SCOPE

15.1 Conclusion

The V-FAB platform provides a functional, web-deployable educational tool that connects theoretical photolithography principles with practical cleanroom operation. By combining custom mechanical modeling, step-by-step interactive workflows, real-time machine learning inference, and fault-tolerant WebGL deployment, the project delivers a reliable and accessible simulation environment for semiconductor engineering education.

15.2 Future Scope

- Stochastic Defect Modeling: Transitioning the PEB and dissolution stages from continuum mathematics to mesoscale stochastic simulations (e.g., Kinetic Monte Carlo). By introducing photon shot noise and discrete molecular reaction-diffusion tracking, the network will be able to predict the probability distribution of stochastic print failures, rather than just average geometric profiles.

- Differentiable Inverse Lithography Technology (ILT): Utilizing the trained forward-predictive digital twin as a differentiable loss layer within an optimization loop. By backpropagating through the optical and chemical U-Nets, the system can automatically synthesize ideal mask patterns, generating advanced Optical Proximity Correction (OPC) features and Sub-Resolution Assist Features (SRAFs) without relying on heuristic rule-based anchoring.

- Virtual Reality (VR) Support: Extending the spatial interaction system to support VR headsets for fully immersive cleanroom operator training.

- Integrity of process steps: Ensuring that a user who performs a step in the wrong order or gets stuck, receives further guidance to complete the tasks in the sequence.

- Real-Time Microscopic 3D Wafer Views: Enhancing the RVE visualizer to support full 3D topological rendering of resist profiles and defect formations during processing.

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